

StructRAG Annotation Schema

1. Purpose and Scope

This annotation schema defines how diagram-based STEM questions are converted into graph-level ground truth for evaluating StructRAG. The schema is designed for diagrams in which the key reasoning target is structural connectivity, including network topologies, circuit-like diagrams, flow or bus structures, hierarchical routing diagrams, and hybrid graph layouts. It supports three tasks: constructing the reference graph, assigning topology labels, and evaluating predicted graphs produced by StructRAG and baseline systems.

The schema intentionally separates structural information from surface-level visual or semantic details. Node labels, component names, numeric values, and drawing layout may be recorded as metadata when useful, but the primary annotation target is the abstract graph structure: which entities exist and which connections are valid.

2. Annotation Unit

The basic annotation unit is one student-submitted diagram-based question. A question may contain text and one or more diagram images. Annotators focus on the diagram image that is required to answer the structural reasoning task. If multiple diagrams are needed jointly, the annotation should specify the primary diagram and any dependent sub-diagram.

Field	Definition	Required
question_id	Unique identifier assigned to each retained question.	Yes
diagram_id	Unique identifier for the diagram image associated with the question.	Yes
discipline	Primary STEM area, such as Electrical Engineering, Computer Science, or Systems Engineering.	Yes
topology_type	Main topology category assigned according to Section 6.	Yes
source_status	Whether the sample belongs to the evaluation dataset, template library, synthetic example set, or pilot set.	Yes
notes	Optional annotator notes on ambiguity, occlusion, or unusual visual conventions.	Optional

3. Graph Representation

Each diagram is represented as a graph $G = (V, E)$, where V is the set of diagram entities and E is the set of valid structural connections. Unless a diagram explicitly uses directional arrows or direction is essential for the question, edges are treated as undirected. When direction is present and instructionally meaningful, the direction should be recorded in the edge metadata.

3.1 Node Annotation

A node is any entity that participates in the structural connectivity of the diagram. Nodes may correspond to labeled points, unlabeled junctions, components, routers, devices, terminals, circuit elements, process blocks, or abstract states. Annotators should include a node only when it is structurally relevant to the graph representation needed for the task.

Node Type	Inclusion Rule	Example
Labeled node	Include when a text label identifies a graph entity.	A, B, R1, u, x
Unlabeled junction	Include when a visual junction creates a meaningful branching or connection point.	Wire intersection that forms a circuit junction
Component node	Include when a component functions as a structural element in the diagram.	Resistor, transistor, router, switch, process block
Terminal or port	Include when the terminal itself is part of the reasoning target.	Input/output port, emitter/base/collector terminal
Decorative mark	Exclude if it does not affect connectivity or reasoning.	Arrow label, legend marker, background grid

3.2 Edge Annotation

An edge represents a valid structural connection between two nodes. Edges may be shown by straight lines, curved lines, wires, buses, arrows, or implicit connections required by domain notation. Annotators should record an edge when the visual evidence or domain convention indicates that two nodes are connected in the structure being assessed.

- Direct visual connection: add an edge when a visible line or wire connects two nodes without interruption.
- Curved or partially occluded connection: add an edge if the intended continuity is clear from visual evidence and domain conventions.
- Crossing lines: do not add a junction unless the diagram convention indicates an actual connection, such as a filled node or explicit junction marker.
- Arrowed connection: record direction only when the arrow represents a directed relation rather than a visual annotation.
- Bus or shared line: represent the bus as edges between the bus node and attached components, or as a bus node plus incident edges when the bus itself is structurally meaningful.

4. Ground-Truth Graph Construction Procedure

Annotators should follow the same sequence for every diagram to ensure consistent graph construction.

1. Inspect the full question and diagram to identify the intended structural reasoning target.
2. List all structurally relevant nodes and assign stable node identifiers.
3. Record all valid edges according to visible connectivity and domain conventions.

4. Mark uncertain visual features in the notes field instead of forcing a connection when evidence is insufficient.
5. Assign one primary topology type and, if needed, one secondary topology descriptor.
6. Export the annotation using the standard JSON or CSV fields defined in Section 10.

5. Handling Ambiguity

Ambiguity is common in scanned, hand-drawn, or visually dense STEM diagrams. Annotators should distinguish between uncertainty in visual parsing and uncertainty in the intended ground truth. If the instructional intent is clear, the ground-truth graph should reflect that intent. If the intent is not clear, the case should be flagged for consensus review.

Ambiguity Type	Annotation Rule
Overlapping labels	Use the label that is visually closest to the node; flag for review if two labels plausibly refer to the same entity.
Curved or broken line	Annotate the edge if continuity and endpoint relation are clear; otherwise mark as uncertain.
Line crossing	Treat as non-connected unless a junction marker or domain convention indicates connection.
Unlabeled component	Assign a generated node identifier and record component_type as unknown or generic.
Multiple possible abstractions	Use the abstraction needed to answer the question; document the alternative in notes.

6. Topology Category Annotation

Each diagram receives one primary topology label. The label is based on abstract connectivity rather than visual layout, node names, or domain-specific labels. The categories are aligned with the template library and experimental analysis.

Topology Type	Definition	Typical Cues
Star	One central node connects directly to multiple peripheral nodes.	Hub-and-spoke structure; high-degree center node
Ring	Nodes form a closed loop, usually with each node connected to two neighbors.	Cycle, loop, redundant path
Chain / Bus	Nodes are arranged in a linear, sequential, or shared-bus structure.	Path-like sequence or common backbone line
Bridge / Mesh	Graph contains redundant cross-connections or multiple alternative paths.	Dense interconnections, bridge links, mesh-like redundancy
Tree	Hierarchical branching structure without cycles.	Root, branches, subnet hierarchy
Hybrid Graph	Combination of multiple recognizable topology patterns.	Star plus ring, bus plus tree, mesh plus hierarchy
Cross-Layer / Layered	Connections occur across explicit layers or levels.	Protocol layers, multi-layer routing, stacked modules
Unclassified / Irregular	Graph does not fit a canonical pattern or contains custom domain-specific structure.	Irregular layout or non-standard connectivity

7. Prediction Evaluation Schema

Predicted graphs are evaluated against the adjudicated ground-truth graph using question-level accuracy and the edge-level F1 score. The goal is to capture both complete-graph correctness and partial edge recovery.

Metric	Decision Rule
Question-level accuracy (Acc)	A prediction is fully correct only if the predicted node set and edge set exactly match the ground-truth topology, with no missing or extra structural elements.
True positive edge (TP)	A predicted edge that appears in the ground-truth graph after canonicalization.
False positive edge (FP)	A predicted edge that does not appear in the ground-truth graph.
False negative edge (FN)	A ground-truth edge that is missing from the predicted graph.
Edge-level F1 score	Computed from edge-level precision and recall using TP, FP, and FN edge counts.

8. Canonicalization Rules Before Evaluation

Before comparing a predicted graph with the ground truth, both graphs are converted into a canonical form. Canonicalization reduces superficial differences that do not affect structural correctness.

- Trim spaces, standardize case when labels are case-insensitive, and normalize common OCR variants when the intended label is clear.
- Sort node lists and edge lists deterministically.
- Treat edges as undirected unless the diagram explicitly encodes direction.
- Represent an undirected edge as an ordered pair using lexical order of node identifiers.
- Remove duplicate edges after canonicalization.
- Do not merge structurally distinct nodes merely because their labels are visually similar.

9. Human Review and Adjudication

The human review process uses a two-stage protocol. Most samples are assigned to one educator panel for primary review, while a stratified reliability subset is reviewed by both panels. Within each panel, assigned samples are first checked independently by two educators. Disagreements are flagged when annotators differ on full-graph correctness or edge-level decisions.

Before consensus adjudication, inter-rater reliability is assessed using Fleiss' kappa for the binary full-correctness judgment. Flagged cases are then resolved through consensus review by re-examining the original diagram, the ground-truth graph, and the predicted graph under the same rubric. The final adjudicated labels are used for reporting Acc and the edge-level F1 score.

10. Recommended Output Format

The following JSON structure is recommended for each annotated diagram. The same schema can be flattened into CSV for spreadsheet-based review.

```
{
  "question_id": "Q0001",
  "diagram_id": "D0001",
  "discipline": "Computer Science",
  "topology_type": "Star",
  "nodes": [
    {"id": "u", "label": "u", "node_type": "labeled_node"},
    {"id": "v", "label": "v", "node_type": "labeled_node"}
  ],
  "edges": [
    {"source": "u", "target": "v", "directed": false, "evidence": "visible_line"}
  ],
  "uncertain_features": [
    {"candidate_edge": ["u", "x"], "reason": "curved or partially occluded line"}
  ],
  "annotation_status": "adjudicated",
  "annotator_notes": "No direct instance overlap with template library."
}
```

11. Quality-Control Checklist

- All structurally relevant nodes are included and have stable identifiers.
- All visible or conventionally implied valid edges are recorded.
- Crossings, junctions, and ambiguous curves are handled according to the schema.
- The topology label reflects abstract connectivity rather than drawing layout.
- Predicted and ground-truth graphs are canonicalized before comparison.
- Ambiguous cases are flagged and resolved through consensus review.
- No personally identifiable information or institution-specific sensitive details are retained in released annotations.